Lecture notes: Mathematical Optimization and Applications

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CHAPTER 1

Linear programming

The aim of this chapter is to introduce linear programs and their duality theory. We start with the convex sets that appear as feasible sets of linear programs, and with Farkas Lemma, which characterizes when a system of linear inequalities has no solution. Parts of this chapter and of Chapter 2 follow [7, Sec. 2.2–2.3]. The chapter contains:

i) notation for vectors, matrices, and the componentwise order (Section 1.1);
ii) convex sets, cones, convex hulls, and polyhedra (Section 1.2);
iii) Farkas Lemma, with the proof of the harder direction in Appendix A (Section 1.3);
iv) primal and dual linear programs, feasibility and optimal values, and how to bring a linear program into standard form (Section 1.4);
v) weak duality and the duality theorem (Section 1.5);
vi) infeasibility certificates (Section 1.6).

1.1. Notation

Throughout, $A \in \mathbb{R}^{m\times n}$ is a real matrix with rows $a_1^T, \dots, a_m^T$,

$$A = \begin{pmatrix} \cdots a_1^T \cdots \\ \vdots \\ \cdots a_m^T \cdots \end{pmatrix}, \tag{1}$$

so that $a_i \in \mathbb{R}^n$. The vectors $b = (b_1, \dots, b_m)^T \in \mathbb{R}^m$ and $c \in \mathbb{R}^n$ are column vectors. The zero vector is $\mathbf{0}$, the all-ones vector is $\mathbf{1}$, and I is the identity matrix.

For $r, s \in \mathbb{R}^m$ we write $r \le s$ if $r_i \le s_i$ for all $i = 1, \dots, m$ (componentwise order). In particular, $r \ge 0$ means $r_i \ge 0$ for all i. For a matrix M, $M \ge 0$ means that all entries of M are non-negative.

Then $Ax \le b$ is the system of m linear inequalities $a_i^T x \le b_i$, $i = 1, \dots, m$. The set of vectors $x \in \mathbb{R}^n$ with $x \ge 0$ is the non-negative orthant $\mathbb{R}^n_+$.

We will use the following property of the componentwise order several times:

$$p \ge 0 \quad \text{and} \quad r \le s \quad \Longrightarrow \quad p^T r \le p^T s\,. \tag{2}$$

Indeed, if $r_i \le s_i$ and $p_i \ge 0$, then $p_i r_i \le p_i s_i$. Summing over all i gives $p^T r = \sum_i p_i r_i \le \sum_i p_i s_i = p^T s$. In words: a non-negative combination of valid linear inequalities is a valid linear inequality.

1.2. Convex sets and polyhedra

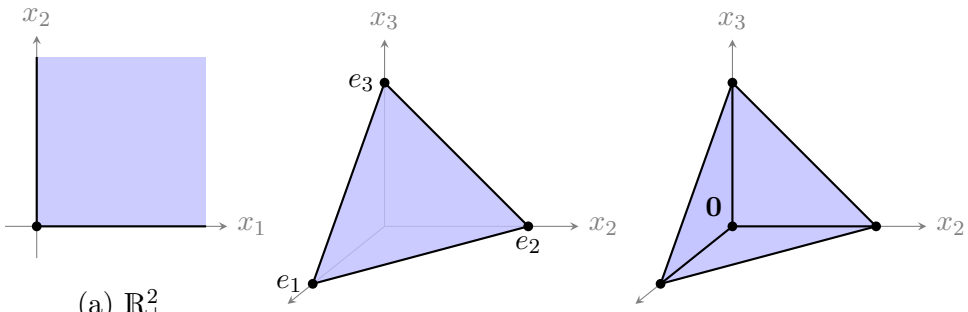
The aim of this section is to introduce the convex sets that appear as feasible sets of linear programs.

1.2.1. Cones. A subset K of a vector space is a cone if $\lambda x \in K$ for all $x \in K$ and $\lambda \ge 0$. See Figure 2(a).

1.2.2. Convex sets. A subset C of a vector space is convex if $\lambda x + (1-\lambda)y \in C$ for all $x, y \in C$ and $\lambda \in [0, 1]$, i.e. if it contains the segment between any two of its points. A convex cone is a cone that is convex. Equivalently, $\lambda x + \mu y \in K$ for all $x, y \in K$ and $\lambda, \mu \ge 0$.

1.2.3. Convex hull. A convex combination of $x_1, \dots, x_k$ is a sum $\sum_i \lambda_i x_i$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$. The convex hull $\operatorname{conv}(S)$ of a set S is the set of all convex combinations of finitely many points of S. It is the smallest convex set containing S. See Figure 2(c), (d).

1.2.4. Polyhedra. A *halfspace* is a set $\{x \in \mathbb{R}^n : a^T x \leq \gamma\}$ with $a \in \mathbb{R}^n$, $a \neq \mathbf{0}$, and $\gamma \in \mathbb{R}$. A *polyhedron* is an intersection of finitely many halfspaces, i.e. a set of the form $\{x \in \mathbb{R}^n : Ax \leq b\}$. A *polytope* is a bounded polyhedron. Every halfspace is convex, and an intersection of convex sets is convex, so every polyhedron is convex. A set is a polytope if and only if it is the convex hull of finitely many points (Minkowski–Weyl theorem, see [11, Ch. 1]).



(a) $\mathbb{R}^2_+$ (b) Δ_3 (c) unit simplex

FIGURE 1. The polyhedra of Example 1 for $n = 2$ and $n = 3$. (a) The non-negative orthant $\mathbb{R}^2_+$ is unbounded, so it is not a polytope. (b) The probability simplex $\Delta_3 = \operatorname{conv}\{e_1, e_2, e_3\}$ is a triangle in $\mathbb{R}^3$. (c) The unit simplex $\operatorname{conv}\{\mathbf{0}, e_1, e_2, e_3\}$ is a tetrahedron.

Example 1. *Some convex sets, see Fig. 1.*

i) The non-negative orthant $\mathbb{R}^n_+ = \{x : -x \leq \mathbf{0}\}$ *is a polyhedron, but not a polytope.*

ii) The probability simplex $\Delta_n = \{x \in \mathbb{R}^n : x \geq 0,\ \mathbf{1}^T x = 1\} = \operatorname{conv}\{e_1, \dots, e_n\}$ *is a polytope.*

iii) The unit simplex $\{x \in \mathbb{R}^n : x \geq 0,\ \mathbf{1}^T x \leq 1\} = \operatorname{conv}\{\mathbf{0}, e_1, \dots, e_n\}$ *is a polytope.*

A *vertex* of a polyhedron Q is a point $v \in Q$ that is not of the form $v = \lambda x + (1-\lambda)z$ with $x, z \in Q$, $x \neq z$, and $\lambda \in (0,1)$.

Fact 2 ([8, Ch. 4], [9, Ch. 8])**.** *Let Q be a non-empty polyhedron that contains no line, for example a polytope or a polyhedron contained in $\mathbb{R}^n_+$. Then Q has a vertex. If a linear objective function has a finite optimal value over Q, then it has an optimal solution that is a vertex of Q.*

To summarize this section: polyhedra are the solution sets of finitely many linear inequalities; they are convex, and the bounded ones (polytopes) are the convex hulls of finitely many points.

1.3. Farkas Lemma

The aim of Farkas Lemma is to completely characterize when a system of linear inequalities $Ax \leq b$ has no solution.

THEOREM 3 (Farkas Lemma, [7, Lemma 2.6]). *Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Exactly one of the following two statements is true:*

i) There exists $x \in \mathbb{R}^n$ such that $Ax \leq b$.

ii) There exist $\lambda_1, \dots, \lambda_m \geq 0$ such that

$$\sum_{i=1}^m \lambda_i a_i = \mathbf{0}\,, \qquad \sum_{i=1}^m \lambda_i b_i = -1\,. \tag{3}$$

Condition ii) can be written in compressed form as $\lambda \in \mathbb{R}^m$, $\lambda \geq 0$ *with* $A^T\lambda = \mathbf{0}$ *and* $b^T\lambda = -1$.

Example 4. *The system* $x \leq 1$, $-x \leq -2$ *in one variable* $x \in \mathbb{R}$ *has no solution, since it asks for* $x \leq 1$ *and* $x \geq 2$. *Here* $a_1 = 1$, $b_1 = 1$, $a_2 = -1$, $b_2 = -2$. *The vector* $\lambda = (1,1)^T$ *satisfies statement ii):* $\lambda_1 a_1 + \lambda_2 a_2 = 1 - 1 = 0$ *and* $\lambda_1 b_1 + \lambda_2 b_2 = 1 - 2 = -1$. *Adding the two inequalities gives* $0 \leq -1$.

This is due to Farkas [2], who stated it in an equivalent form. The version above and other variants are in [8, Ch. 6].

Statement ii) says: the non-negative combination of the inequalities $a_i^T x \leq b_i$ with weights λ_i yields the inequality $0 \leq -1$, which is a contradiction. Thus no $x \in \mathbb{R}^n$ satisfying $Ax \leq b$ exists. We can think of λ as a short proof that i) fails [7, Sec. 2.2]. Farkas lemma says that such a proof exists whenever i) fails, and vice versa.

Proof that ii) excludes i). Suppose λ satisfies the constraints (3) and x satisfies $Ax \leq b$. By (2) with $p = \lambda \geq 0$,

$$\lambda^T A x \leq \lambda^T b\,. \tag{4}$$

But the left-hand side equals $(A^T\lambda)^T x = \mathbf{0}^T x = 0$, whereas the right-hand side equals $\lambda^T b = -1$. This is a contradiction. □

The other direction, that ii) holds whenever i) fails, is harder. The proof of this direction uses Fourier–Motzkin elimination and is given in Appendix A. It will not appear in the exam.

To summarize this section: the system $Ax \leq b$ has no solution if and only if some non-negative combination of its inequalities yields the inequality $0 \leq -1$.

1.4. Linear programming

1.4.1. Primal and dual LPs. The aim of this subsection is to define the primal and dual linear programs that we use throughout.

primal:

$$\text{(P)} \qquad \begin{aligned} &\max && c^T x \\ &\text{subject to} && Ax \leq b \end{aligned}$$

Here the matrix $A \in \mathbb{R}^{m\times n}$ and the vectors $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ are the parameters, and $x \in \mathbb{R}^n$ is the vector of variables. In components, the constraints are $a_i^T x \leq b_i$ for $i = 1, \ldots, m$.

To every primal linear program one associates a dual program with the same parameters A, b, c [7, Sec. 2.2].

dual:

$$\text{(D)} \qquad \begin{aligned} &\min && b^T y \\ &\text{subject to} && A^T y = c \\ & && y \geq 0 \end{aligned}$$

Here $y \in \mathbb{R}^m$ is the vector of variables.

Note that the primal program has n variables and m inequality constraints. In contrast, the dual has m variables (one for each primal constraint), n equality constraints (one for each primal variable), and m sign constraints $y_i \geq 0$.

The objective value of every feasible point of the dual is an upper bound on the objective value of every feasible point of the primal. This is weak duality, proven in Section 1.5.

1.4.2. Feasibility and optimal values. The aim of this subsection is to fix the terms used to describe the outcome of a linear program.

Definition 5. *For the primal program* (P)*:*

i) *A* feasible point *(or feasible solution) is a vector* $x \in \mathbb{R}^n$ *with* $Ax \leq b$.
ii) *The* feasible set *is* $\mathcal{F}_P = \{x \in \mathbb{R}^n : Ax \leq b\}$.
iii) *The program is* feasible *if* $\mathcal{F}_P \neq \emptyset$*, and* infeasible *otherwise.*
iv) *The* optimal value *is* $\alpha = \sup\{c^T x : x \in \mathcal{F}_P\}$*, with the convention* $\alpha = -\infty$ *if the program is infeasible.*
v) *The program is* unbounded *if* $\alpha = +\infty$*, i.e. if for every* $M \in \mathbb{R}$ *there is a feasible* x *with* $c^T x > M$.
vi) *An* optimal solution *is a feasible point* x^* *with* $c^T x^* = \alpha$.

For the dual program (D)*, the feasible set is* $\mathcal{F}_D = \{y \in \mathbb{R}^m : A^T y = c,\ y \geq 0\}$*, the optimal value is* $\beta = \inf\{b^T y : y \in \mathcal{F}_D\}$ *with the convention* $\beta = +\infty$ *if the program is infeasible, the program is unbounded if* $\beta = -\infty$*, and an optimal solution is a feasible point* y^* *with* $b^T y^* = \beta$.

The conventions $\alpha = -\infty$ and $\beta = +\infty$ for infeasible programs make the inequality $\alpha \leq \beta$ of Section 1.5 hold in all cases.

The feasible set $\mathcal{F}_P$ is a polyhedron (Section 1.2). So is $\mathcal{F}_D$, since an equality is a pair of inequalities (Section 1.4.3).

1.4.3. Transforming linear programs into standard form. The aim of this subsection is to bring any linear program into the primal standard form (P). Its dual is then (D).

There are many different conventions for linear programs (and also semidefinite programs), using min or max, constraints of type $\leq$, $\geq$, $=$, and sign constraints on the variables. These can be converted into each other [7, Sec. 2.2]. Instead of memorizing a dual for each convention, we convert to (P) and dualize from there. Four rules suffice:

i) Turn minimization into maximization: $\min\{c^T x : x \in \mathcal{F}\} = -\max\{(-c)^T x : x \in \mathcal{F}\}$, and both programs have the same optimal solutions.
ii) Reverse inequalities by $a^T x \geq \gamma \iff -a^T x \leq -\gamma$.
iii) Model equalities as pairs of inequalities:

$$Ax = b \quad \Longleftrightarrow \quad Ax \leq b \quad \text{and} \quad -Ax \leq -b\,. \tag{5}$$

iv) Sign constraints as inequalities: $x \geq 0 \iff -Ix \leq \mathbf{0}$.

In (P) the variables are not restricted in sign, so no further rules are needed.

Two further transformations go in the other direction, towards the form of (D) with equality constraints and non-negative variables. Solvers often expect one of these forms.

i) Inequalities by equalities with a slack variable z:

$$a^T x \leq \gamma \quad \Longleftrightarrow \quad \text{there is } z \geq 0 \text{ with } a^T x + z = \gamma\,. \tag{6}$$

In matrix form, $Ax \leq b$ if and only if there is $z \in \mathbb{R}^m$, $z \geq 0$, with

$$\begin{pmatrix} A & I \end{pmatrix} \cdot \begin{pmatrix} x \\ z \end{pmatrix} = b\,. \tag{7}$$

ii) Free variables by two non-negative variables: every $x \in \mathbb{R}^n$ can be written as

$$x = x_+ - x_- \quad \text{with} \quad x_+, x_- \in \mathbb{R}^n_+\,. \tag{8}$$

So inequality constraints can be exchanged for equality constraints together with additional non-negative variables, and free variables can be exchanged for twice as many non-negative variables.

Example 6. *Consider*

$$\begin{aligned} \min \quad & x_1 + x_2 + x_3 \\ \textit{subject to} \quad & x_1 - x_2 = 0 \\ & x_1 + x_3 \leq 1\,. \end{aligned} \tag{9}$$

By rule i), its optimal value is $-\alpha$, where $\alpha = \max\{-x_1 - x_2 - x_3 : x_1 - x_2 = 0,\ x_1 + x_3 \leq 1\}$.

With rule iii) we can rewrite the constraints as $x_1 - x_2 \leq 0$, $-x_1 + x_2 \leq 0$, $x_1 + x_3 \leq 1$. This yields a primal program in standard form with

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad c = \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}. \tag{10}$$

The dual (D) is

$$\begin{aligned} \min \quad & y_3 \\ \textit{subject to} \quad & y_1 - y_2 + y_3 = -1 \\ & -y_1 + y_2 = -1 \\ & y_3 = -1 \\ & y \geq 0\,. \end{aligned} \tag{11}$$

Note that the constraint $y_3 = -1$ contradicts $y_3 \geq 0$, and so the dual is infeasible. By the duality theorem (Section 1.5.2), the primal has no finite optimal value, so it is unbounded or infeasible. It is feasible ($x = \mathbf{0}$), so it is unbounded: the points $x = (-t, -t, 0)$ with $t \geq 0$ are feasible with $-x_1 - x_2 - x_3 = 2t \to \infty$. The original minimization problem has optimal value $-\infty$.

1.5. Duality theory

The aim of this section is to show that dual objective values are upper bounds on primal objective values (weak duality), and to state when the optimal values are equal (duality theorem).

1.5.1. Weak duality.

Proposition 7 (Weak duality)**.** *For every pair of feasible solutions (x, y) of the primal and dual program, it holds that*

$$c^T x \leq b^T y\,. \tag{12}$$

Proof. Use the primal and dual constraints and (2) with $p = y \geq 0$ and $Ax \leq b$:

$$c^T x = (A^T y)^T x = y^T A x \leq y^T b = b^T y\,. \tag{13}$$

□

i) Taking the supremum over x and the infimum over y gives $\alpha \leq \beta$. With the conventions of Section 1.4.2, this also holds if one of the programs is infeasible.

ii) For any pair of feasible solutions (x, y) and optimal solutions (x^*, y^*),

$$c^T x \leq c^T x^* = \alpha \leq \beta = b^T y^* \leq b^T y\,. \tag{14}$$

So the objective value of a feasible x is a lower bound on α, and the objective value of a feasible y is an upper bound on α and on β.

iii) If $c^T x = b^T y$ for a feasible pair (x, y), then x and y are optimal solutions.

iv) If the primal is unbounded, $\alpha = +\infty$, then $\beta = +\infty$, so the dual is infeasible. If the dual is unbounded, the primal is infeasible.

v) The *duality gap* is $\Delta = \beta - \alpha \geq 0$.

vi) *Strong duality* holds if $\Delta = 0$, i.e. $\alpha = \beta$.

See Fig. 2.

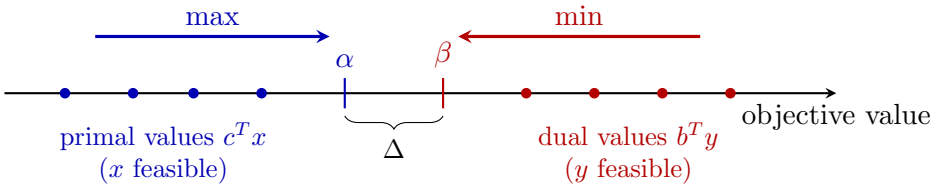


FIGURE 2. Weak duality. The maximization pushes the primal objective values $c^T x$ up from below, and the minimization pushes the dual objective values $b^T y$ down from above. Every primal value is at most every dual value, so $\alpha \leq \beta$, with duality gap $\Delta = \beta - \alpha \geq 0$. For linear programs with a finite optimal value, the duality theorem (Theorem 8) gives $\Delta = 0$.

1.5.2. Duality theorem. We now state the duality theorem and list which combinations of primal and dual outcomes can occur.

THEOREM 8 (Duality theorem, [7, Thm. 2.8], [8, Ch. 6]). *If either the primal or the dual program has a finite optimal value, then both programs have optimal solutions, and their optimal values are equal, $\alpha = \beta$.*

We will not discuss the proof here. It can be derived from Farkas Lemma, see [8, Ch. 6]. Together with weak duality, the duality theorem gives two facts about a primal-dual pair:

i) If the primal or the dual has a finite optimal value, then so does the other, both are attained, and they are equal. (Duality theorem.)
ii) If the primal or the dual is unbounded, then the other is infeasible. (Weak duality.)

The remaining cases follow from these two facts. For example, if the primal is infeasible, then by fact i) the dual has no finite optimal value, so the dual is unbounded or infeasible. Both cases can occur. The following examples show the four combinations that are possible:

i) Both have optimal solutions: max x_1 s.t. $x_1 \leq 1$. Dual: min y_1 s.t. $y_1 = 1$, $y_1 \geq 0$. Here $\alpha = \beta = 1$.
ii) Primal unbounded, dual infeasible: max x_1 s.t. $-x_1 \leq 0$. Dual: min 0 s.t. $-y_1 = 1$, $y_1 \geq 0$.
iii) Primal infeasible, dual unbounded: max x_1 s.t. $x_1 \leq -1$, $-x_1 \leq -1$. Dual: min $(-y_1 - y_2)$ s.t. $y_1 - y_2 = 1$, $y \geq 0$. The points $y = (1 + t, t)$ with $t \geq 0$ are feasible, and their objective value $-1 - 2t$ tends to $-\infty$.
iv) Both infeasible: max x_2 s.t. $x_1 \leq -1$, $-x_1 \leq -1$. The primal requires $x_1 \leq -1$ and $x_1 \geq 1$. Dual: min $(-y_1 - y_2)$ s.t. $y_1 - y_2 = 0$, $0 = 1$, $y \geq 0$. The second dual constraint (for the variable x_2) cannot be satisfied.

Remark 9. *As we will see, the duality theorem does not hold for semidefinite programs: there, the optimal values of primal and dual can differ, and a finite optimal value need not be attained.*

1.6. Infeasibility certificates

We now show how a vector $y \geq 0$ with $A^T y = \mathbf{0}$ and $b^T y < 0$ proves that the primal program is infeasible. For the feasibility problem ($c = 0$), such y is a dual feasible point with negative objective value. For general c, it need not be dual feasible. For this we will use Farkas lemma!

Let $y \in \mathbb{R}^m$ with $y \geq 0$, $A^T y = \mathbf{0}$, and $b^T y = -1$. Then the primal problem is infeasible. Such y is called an *infeasibility certificate*. In particular, it acts as an instance λ of Farkas Lemma (Theorem 3). Conversely, by Farkas Lemma (Theorem 3), every infeasible primal program has an infeasibility certificate.

i) For the feasibility problem ($c = 0$), such y is a feasible dual point with $b^T y = -1$. Then Cy is dual feasible for any $C > 0$, and $b^T(Cy) = -C \to -\infty$. So the dual is unbounded. By weak duality, any dual objective value is an upper bound on any primal objective value. This pushes the primal optimum to $-\infty$, and thus the primal problem is infeasible.

ii) For general c, if y_0 is dual feasible, then so is $y_0 + Cy$ for any $C > 0$, and $b^T(y_0 + Cy) = b^T y_0 - C \to -\infty$. The same argument applies.
iii) In case iii) of Section 1.5.2, $y = (1/2, 1/2)$ is an infeasibility certificate.

1.7. Summary

To summarize this chapter:

i) The feasible sets of linear programs are polyhedra, which are convex.
ii) Farkas Lemma: the system $Ax \leq b$ has no solution if and only if some non-negative combination of its inequalities is the inequality $0 \leq -1$.
iii) Every linear program can be brought into the standard form (P), which has the dual (D).
iv) Weak duality: dual objective values are upper bounds on primal objective values.
v) Duality theorem: if the primal or the dual has a finite optimal value, then both optimal values are attained and equal. If the primal or the dual is unbounded, then the other is infeasible.
vi) An infeasible primal has an infeasibility certificate $y \geq 0$ with $A^T y = \mathbf{0}$ and $b^T y = -1$.

CHAPTER 2

Linear programs in combinatorial optimization

The aim of this chapter is to use linear programs to bound graph parameters. For each problem we write it as an integer linear program, relax it to a linear program, and compare the two optimal values. This approach is described in [7, Sec. 1 and 2.3]. The chapter contains:

i) integer linear programs and LP relaxations (Section 2.2);
ii) stable sets and the polytopes $\mathrm{STAB}(G) \subseteq \mathrm{QSTAB}(G) \subseteq \mathrm{FSTAB}(G)$ (Section 2.3);
iii) matchings and vertex covers, whose LP relaxations are dual to each other, and König's theorem (stated without proof) (Section 2.4);
iv) flows from a source to a target, and the maximum flow problem as a linear program (Section 2.5).

2.1. Graphs

A graph $G = (V, E)$ consists of a finite set V of vertices and a set E of edges $e = \{v, w\}$ with $v, w \in V$, $v \neq w$. For $v \in V$, $\delta(v)$ is the set of edges that contain v. For $U \subseteq V$, $E(U)$ is the set of edges with both endpoints in U. A graph is *bipartite* if $V = V_1 \cup V_2$ with $V_1 \cap V_2 = \emptyset$ such that every edge has one endpoint in V_1 and one in V_2. A *clique* is a set of pairwise adjacent vertices. K_n is the complete graph on n vertices; K_3 is the triangle.

We index vectors by vertices or edges: $x \in \mathbb{R}^V$ or $x \in \mathbb{R}^E$. For $S \subseteq V$, the incidence vector $\chi^S \in \{0,1\}^V$ has $\chi^S_v = 1$ if $v \in S$ and 0 otherwise; similarly $\chi^F \in \{0,1\}^E$ for $F \subseteq E$. For $x \in \mathbb{R}^V$ and $S \subseteq V$ we write $x(S) = \sum_{v \in S} x_v$, and similarly for edges. The vertex-edge incidence matrix $B \in \{0,1\}^{V \times E}$ has $B_{v,e} = 1$ if $v \in e$ and 0 otherwise. Then $(Bx)_v = x(\delta(v))$ for $x \in \mathbb{R}^E$, and $(B^T y)_e = y_v + y_w$ for $y \in \mathbb{R}^V$ and $e = \{v, w\}$.

2.2. Integer linear programs and LP relaxations

An integer linear program is

$$\begin{array}{lll} \max & c^T x & \\ \text{subject to} & Ax \leq b\,, & x \in \mathbb{Z}^n\,. \end{array} \tag{15}$$

Solving integer linear programs is NP-hard in general [5]. Its *LP relaxation* is the linear program (P) obtained by dropping the constraint $x \in \mathbb{Z}^n$. The feasible set of the relaxation contains the feasible set of the integer program. So the optimal value of the LP relaxation is an upper bound on the optimal value of the integer program (for minimization problems, a lower bound).

For some problems the LP relaxation has an integral optimal solution, and then the LP relaxation and the integer program have the same optimal value. Below we see examples: stable sets and matchings in bipartite graphs, and flows.

2.3. Stable sets

A set $S \subseteq V$ is *stable* (or independent) if no edge has both endpoints in S. The stability number $\alpha(G)$ is the largest size of a stable set. See Fig. 1.

Computing $\alpha(G)$ is NP-hard [5], see also [7, Sec. 2.3]. As an integer linear program,

$$\alpha(G) = \max\{\mathbf{1}^T x : x_v + x_w \leq 1 \text{ for all } \{v, w\} \in E,\ 0 \leq x \leq \mathbf{1},\ x \in \mathbb{Z}^V\}\,. \tag{16}$$

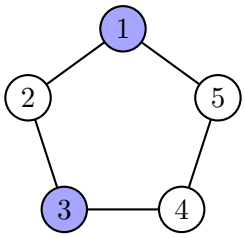


FIGURE 1. The 5-cycle C_5. The filled vertices form the stable set $S = \{1, 3\}$: no edge joins two filled vertices. Every set of three vertices of C_5 contains an edge, so $\alpha(C_5) = 2$.

The feasible integer points are exactly the incidence vectors χ^S of stable sets S. We consider three polytopes:

$$\begin{aligned} \mathrm{STAB}(G) &= \mathrm{conv}\{\chi^S : S \subseteq V \text{ stable}\}\,, \\ \mathrm{QSTAB}(G) &= \{x \in \mathbb{R}^V : x \geq 0,\ x(K) \leq 1 \text{ for all cliques } K\}\,, \\ \mathrm{FSTAB}(G) &= \{x \in \mathbb{R}^V : 0 \leq x \leq \mathbf{1},\ x_v + x_w \leq 1 \text{ for all } \{v, w\} \in E\}\,. \end{aligned} \tag{17}$$

$\mathrm{STAB}(G)$ is the stable set polytope, $\mathrm{FSTAB}(G)$ is the feasible set of the LP relaxation above, and $\mathrm{QSTAB}(G)$ uses clique inequalities instead of edge inequalities. These polytopes are defined as in [7, Sec. 2.3]. There, $\mathrm{FSTAB}(G)$ has no upper bounds $x \leq \mathbf{1}$, since graphs are assumed to have no isolated vertices.

We have

$$\mathrm{STAB}(G) \subseteq \mathrm{QSTAB}(G) \subseteq \mathrm{FSTAB}(G)\,. \tag{18}$$

The first inclusion holds since a stable set meets every clique in at most one vertex, so $\chi^S \in \mathrm{QSTAB}(G)$, and $\mathrm{QSTAB}(G)$ is convex. The second inclusion holds since single vertices and edges are cliques. Maximizing $\mathbf{1}^T x$ over the three sets gives

$$\alpha(G) \leq \max_{x \in \mathrm{QSTAB}(G)} \mathbf{1}^T x \leq \max_{x \in \mathrm{FSTAB}(G)} \mathbf{1}^T x\,. \tag{19}$$

So both LPs give upper bounds on $\alpha(G)$, and the LP over $\mathrm{QSTAB}(G)$ gives an upper bound that is at most the one from $\mathrm{FSTAB}(G)$.

Known results:

i) $\mathrm{STAB}(G) = \mathrm{FSTAB}(G)$ if and only if G is bipartite [7, Prop. 2.10], [4, Ch. 9].
ii) Optimizing a linear function over $\mathrm{QSTAB}(G)$ is NP-hard in general [4, Ch. 9]. The Lovász ϑ function gives a convex set between $\mathrm{STAB}(G)$ and $\mathrm{QSTAB}(G)$ over which one can optimize efficiently [4, Ch. 9].

Example 10. *See Fig. 2 for the first two cases.*

i) Triangle K_3. The stable sets are $\emptyset$, $\{1\}$, $\{2\}$, $\{3\}$, so $\alpha(K_3) = 1$ and $\mathrm{STAB}(K_3) = \mathrm{conv}\{\mathbf{0}, e_1, e_2, e_3\}$, the unit simplex. The clique $\{1, 2, 3\}$ gives

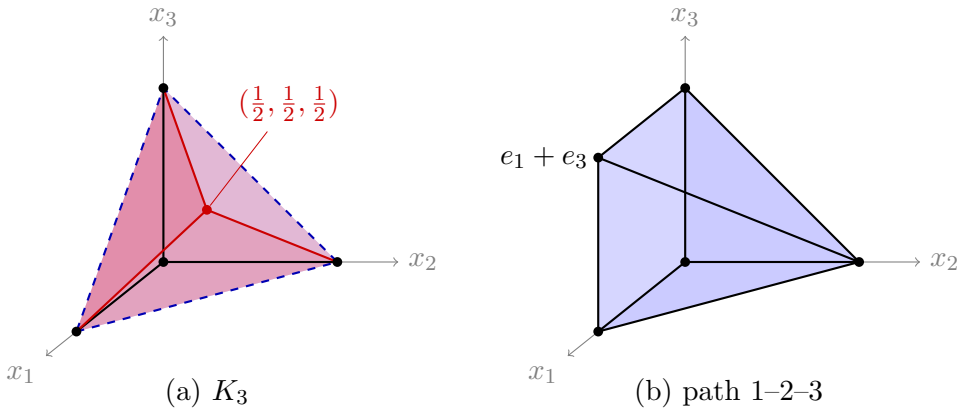
$$\mathrm{QSTAB}(K_3) = \{x \geq 0 : x_1 + x_2 + x_3 \leq 1\} = \mathrm{STAB}(K_3)\,. \tag{20}$$

But $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}) \in \mathrm{FSTAB}(K_3)$, so $\mathrm{FSTAB}(K_3) \neq \mathrm{STAB}(K_3)$, see [7, Fig. 5]. Summing the three edge inequalities gives $2(x_1 + x_2 + x_3) \leq 3$, so $\max_{x \in \mathrm{FSTAB}(K_3)} \mathbf{1}^T x = \frac{3}{2}$.

ii) The path with vertices $1, 2, 3$ and edges $\{1, 2\}$, $\{2, 3\}$. The stable sets are $\emptyset$, $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 3\}$, so $\alpha = 2$. The graph is bipartite, and

$$\mathrm{STAB} = \mathrm{QSTAB} = \mathrm{FSTAB} = \{x : 0 \leq x \leq \mathbf{1},\ x_1 + x_2 \leq 1,\ x_2 + x_3 \leq 1\}\,. \tag{21}$$

iii) Complete graph K_n, $n \geq 2$. Here $\alpha(K_n) = 1$, and $\max_{x \in \mathrm{QSTAB}(K_n)} \mathbf{1}^T x = 1$. But $\frac{1}{2}\mathbf{1} \in \mathrm{FSTAB}(K_n)$, so $\max_{x \in \mathrm{FSTAB}(K_n)} \mathbf{1}^T x \geq n/2$. So the upper bound on $\alpha(G)$ from $\mathrm{FSTAB}(G)$ can exceed $\alpha(G)$ by a factor of $n/2$.



(a) K_3 (b) path 1–2–3

FIGURE 2. Black dots: incidence vectors of stable sets. (a) For the triangle K_3, $\mathrm{STAB}(K_3) = \mathrm{QSTAB}(K_3)$ is the simplex $\mathrm{conv}\{\mathbf{0}, e_1, e_2, e_3\}$ (blue). $\mathrm{FSTAB}(K_3)$ additionally contains the pyramid over the face $\mathrm{conv}\{e_1, e_2, e_3\}$ (dashed) with apex $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ (red), cf. [7, Fig. 5]. (b) For the path with edges $\{1, 2\}$, $\{2, 3\}$, $\mathrm{STAB} = \mathrm{QSTAB} = \mathrm{FSTAB} = \mathrm{conv}\{\mathbf{0}, e_1, e_2, e_3, e_1 + e_3\}$.

To summarize this section: the LPs over $\mathrm{QSTAB}(G)$ and $\mathrm{FSTAB}(G)$ give upper bounds on the stability number $\alpha(G)$. For bipartite graphs the upper bound from $\mathrm{FSTAB}(G)$ equals $\alpha(G)$; for K_n it is $n/2$, while $\alpha(K_n) = 1$.

2.4. Matchings and vertex covers

The aim of this section is to compare the matching number and the vertex cover number of a graph, using that their LP relaxations are dual to each other.

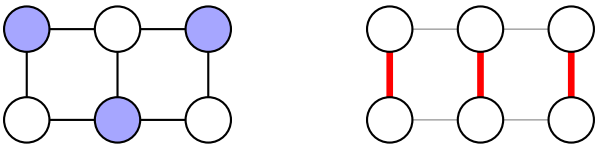
2.4.1. Matchings. A *matching* is a set $M \subseteq E$ of edges such that no two edges in M share a vertex. The matching number $\nu(G)$ is the largest size of a matching. As an integer linear program,

$$\nu(G) = \max\{\mathbf{1}^T x : Bx \le \mathbf{1},\ x \ge 0,\ x \in \mathbb{Z}^E\}\,. \tag{22}$$

Here $Bx \le \mathbf{1}$ says $x(\delta(v)) \le 1$ for every vertex. The feasible integer x are the incidence vectors of matchings.

A matching is *perfect* if it covers every vertex. A maximum weight perfect matching for weights $w \in \mathbb{R}^E$ is an optimal integer solution of $\max\{w^T x : Bx = \mathbf{1},\ x \ge 0\}$. For bipartite graphs this LP has an integral optimal solution whenever it has an optimal solution [**10**, Ch. 18].

2.4.2. Vertex covers. A *vertex cover* is a set $C \subseteq V$ such that every edge has at least one endpoint in C. The vertex cover number $\tau(G)$ is the smallest size of a vertex cover. See Fig. 3 for a vertex cover and a matching.



(a) vertex cover (b) matching

FIGURE 3. A bipartite graph with 6 vertices and 7 edges. (a) The filled vertices form a vertex cover: every edge has at least one filled endpoint, i.e. every edge is "seen" by a filled vertex. (b) The thick edges form a matching: they pair up vertices, and no two of them share a vertex. Here $\nu(G) = \tau(G) = 3$.

As an integer linear program,

$$\tau(G) = \min\{\mathbf{1}^T y : B^T y \ge \mathbf{1},\ y \ge 0,\ y \in \mathbb{Z}^V\}\,. \tag{23}$$

Here $B^T y \ge \mathbf{1}$ says $y_v + y_w \ge 1$ for every edge $\{v, w\}$. An optimal integer solution has entries in $\{0, 1\}$ (replacing entries larger than 1 by 1 keeps feasibility), and is the incidence vector of a vertex cover.

Since the edges of a matching are disjoint, each vertex of a vertex cover covers at most one edge of a matching. Hence $\nu(G) \le \tau(G)$.

2.4.3. Duality of the LP relaxations. Let $\nu^*(G)$ and $\tau^*(G)$ be the optimal values of the LP relaxations of the two integer programs above. We show that the two LP relaxations are dual to each other, using the recipe of Section 1.4.3. In standard form (P), the matching relaxation is

$$\max\ \mathbf{1}^T x \quad \text{s.t.} \quad \begin{pmatrix} B \\ -I \end{pmatrix} x \le \begin{pmatrix} \mathbf{1} \\ \mathbf{0} \end{pmatrix}. \tag{24}$$

Its dual (D) has variables $y \in \mathbb{R}^V$ and $s \in \mathbb{R}^E$:

$$\min\ \mathbf{1}^T y + \mathbf{0}^T s \quad \text{s.t.} \quad B^T y - s = \mathbf{1}\,, \quad y \ge 0\,, \quad s \ge 0\,. \tag{25}$$

Since $s = B^T y - \mathbf{1}$, the constraints on s say $B^T y \ge \mathbf{1}$. So the dual is the vertex cover relaxation. Both programs are feasible ($x = \mathbf{0}$, $y = \mathbf{1}$), so by weak duality and Theorem 8,

$$\nu(G) \le \nu^*(G) = \tau^*(G) \le \tau(G)\,. \tag{26}$$

Example 11. *For the triangle K_3: $\nu(K_3) = 1$ and $\tau(K_3) = 2$. The points $x = \frac{1}{2}\mathbf{1} \in \mathbb{R}^E$ and $y = \frac{1}{2}\mathbf{1} \in \mathbb{R}^V$ are feasible for the two relaxations with the same objective value $\frac{3}{2}$. By weak duality, both are optimal, and $\nu^*(K_3) = \tau^*(K_3) = \frac{3}{2}$.*

THEOREM 12 (König [6]). *If G is bipartite, then $\nu(G) = \nu^*(G) = \tau^*(G) = \tau(G)$.*

We do not prove this here; see [10, Ch. 16].

2.4.4. Edmonds' characterization of the matching polytope. For general graphs, the matching relaxation is not integral, as the triangle shows. Edmonds [1] showed that the matching polytope $\operatorname{conv}\{\chi^M : M \text{ matching}\}$ equals

$$\left\{x \in \mathbb{R}^E : x \geq 0,\ Bx \leq \mathbf{1},\ x(E(U)) \leq \tfrac{|U|-1}{2} \text{ for all } U \subseteq V \text{ with } |U| \text{ odd}\right\}. \tag{27}$$

We do not prove this here. The inequalities $x(E(U)) \leq \frac{|U|-1}{2}$ are examples of Chvátal–Gomory cuts (Section 2.6).

To summarize this section: the matching number and the vertex cover number satisfy $\nu(G) \leq \nu^*(G) = \tau^*(G) \leq \tau(G)$, where the middle equality is LP duality. For bipartite graphs all four numbers are equal.

2.5. Flows

The aim of this section is to write the maximum flow problem as a linear program.

A directed graph $D = (V, E)$ has a set of directed edges $E \subseteq V \times V$; the directed edge $e = (v, w)$ goes from v to w. For $v \in V$, $\delta^{\text{out}}(v)$ is the set of directed edges leaving v, and $\delta^{\text{in}}(v)$ is the set of directed edges entering v. Fix two vertices, a *source* s and a *target* (or *sink*) $t \neq s$, and capacities $\kappa \in \mathbb{R}^E_+$ on the directed edges.

An *s-t flow* is a vector $f \in \mathbb{R}^E$ with

$$\begin{aligned} 0 \leq f_e \leq \kappa_e \quad &\text{for all } e \in E\,, \\ f(\delta^{\text{in}}(v)) = f(\delta^{\text{out}}(v)) \quad &\text{for all } v \in V \setminus \{s, t\}\,. \end{aligned} \tag{28}$$

The first condition says that the flow on each directed edge is non-negative and at most its capacity. The second condition (flow conservation) says that at every vertex other than source and target, the flow entering equals the flow leaving. The value of the flow is the net flow leaving the source, $\operatorname{val}(f) = f(\delta^{\text{out}}(s)) - f(\delta^{\text{in}}(s))$. See Fig. 4.

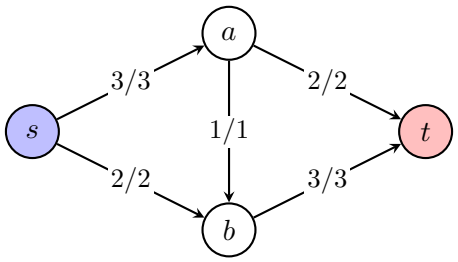


FIGURE 4. A flow from the source s to the target t. Each directed edge e is labelled f_e/κ_e (flow/capacity). Flow conservation holds at a (in: 3, out: $1+2$) and at b (in: $2+1$, out: 3). The value of the flow is $3+2=5$. No flow has larger value, since the two directed edges leaving s have total capacity 5.

The maximum flow problem is the linear program

$$\max\ \operatorname{val}(f) \quad \text{s.t.} \quad f \text{ is an } s\text{-}t \text{ flow}\,. \tag{29}$$

It is feasible ($f = \mathbf{0}$) and bounded ($\operatorname{val}(f) \leq \kappa(\delta^{\text{out}}(s))$), so by Theorem 8 it has an optimal solution. If all capacities are integers, then there is a maximum flow with integer values [3].

Exercise 13. *Bring the maximum flow problem into the standard form (P) and compute its dual.*

To summarize this section: a flow from the source to the target respects the capacities and flow conservation, and maximizing its value is a linear program.

2.6. Chvátal–Gomory cuts

Placeholder: Chvátal–Gomory cuts and Chvátal rank. To be written.

2.7. Summary

To summarize this chapter:

i) The LP relaxation of an integer program gives an upper bound on the optimal value of the integer program for maximization problems, and a lower bound for minimization problems.

ii) Stable sets: $\mathrm{STAB}(G) \subseteq \mathrm{QSTAB}(G) \subseteq \mathrm{FSTAB}(G)$, so LPs over $\mathrm{QSTAB}(G)$ and $\mathrm{FSTAB}(G)$ give upper bounds on $\alpha(G)$. $\mathrm{STAB}(G) = \mathrm{FSTAB}(G)$ if and only if G is bipartite.

iii) Matchings and vertex covers: the two LP relaxations are dual to each other, so $\nu(G) \leq \nu^*(G) = \tau^*(G) \leq \tau(G)$. For bipartite graphs all four numbers are equal (König's theorem). For general graphs, Edmonds' odd set inequalities describe the matching polytope.

iv) Flows: the maximum flow problem is a linear program. For integer capacities it has an optimal solution with integer values.

CHAPTER 3

Review linear algebra

The aim of this chapter is to review the linear algebra used in the other chapters, in particular for positive semidefinite matrices. We generally consider real vector spaces, except for a few examples on Pauli matrices and qubits. The chapter contains:

i) vector spaces, bases, and dimension (Section 3.1);
ii) vectors: inner product, outer product, and expansion in an orthonormal basis (Section 3.2);
iii) matrices: products, trace, trace inner product, and projectors (Section 3.3);
iv) eigenvalues, eigenvectors, and the spectral theorem (Section 3.4);
v) positive semidefinite matrices and equivalent definitions (Section 3.5);
vi) the Löwner order, the PSD cone, and dual cones (Section 3.6).

3.1. Vector spaces

3.1.1. Definition. A real vector space is a set V with an addition $+\colon V \times V \to V$ and a scalar multiplication $\cdot\colon \mathbb{R} \times V \to V$ such that for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$:

$$u + (v + w) = (u + v) + w, \qquad \alpha(v + w) = \alpha v + \alpha w, \tag{30}$$

$$v + w = w + v, \qquad (\alpha + \beta)v = \alpha v + \beta v, \tag{31}$$

$$\text{there is } 0 \in V \text{ with } v + 0 = v, \qquad \alpha(\beta v) = (\alpha\beta)v, \tag{32}$$

$$\text{there is } -v \in V \text{ with } v + (-v) = 0, \qquad 1 \cdot v = v. \tag{33}$$

Example 14. *The spaces $\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$ with entrywise addition and scalar multiplication are real vector spaces.*

A complex vector space is defined in the same way, with $\mathbb{C}$ in place of $\mathbb{R}$.

3.1.2. Basis and dimension. Vectors $v_1, \ldots, v_k \in V$ are linearly independent if $\sum_i \alpha_i v_i = 0$ implies $\alpha_1 = \cdots = \alpha_k = 0$. Their span is the set of all linear combinations $\sum_i \alpha_i v_i$. A basis of V is a linearly independent set that spans V. Every $v \in V$ is then a unique linear combination of the basis vectors. All bases of a finite-dimensional space have the same size, called the dimension $\dim V$. We have $\dim \mathbb{R}^n = n$ and $\dim \mathbb{R}^{m\times n} = mn$.

3.2. Vectors

3.2.1. Notation. A vector $v \in \mathbb{R}^n$ is a column with entries v_i:

$$v = (v_1, \ldots, v_n) = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, \qquad v^T = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}. \tag{34}$$

3.2.2. Inner product. The standard inner product on $\mathbb{R}^n$ is

$$\langle v, w \rangle = v^T w = \sum_{i=1}^{n} v_i w_i. \tag{35}$$

For all $u, v, w \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$:

$$\langle v, w\rangle = \langle w, v\rangle \qquad \text{(symmetry)}, \tag{36}$$

$$\langle u, \alpha v + \beta w\rangle = \alpha\langle u, v\rangle + \beta\langle u, w\rangle \qquad \text{(linearity)}, \tag{37}$$

$$\langle v, v\rangle \geq 0, \quad \langle v, v\rangle = 0 \iff v = 0 \qquad \text{(positive definiteness)}. \tag{38}$$

The induced norm is $\|v\| = \sqrt{\langle v, v\rangle}$. Vectors v, w are orthogonal if $\langle v, w\rangle = 0$.

3.2.3. Outer product. For $v \in \mathbb{R}^m$ and $w \in \mathbb{R}^n$, the outer product is the $m \times n$ matrix

$$vw^T = (v_i w_j)_{ij}\,. \tag{39}$$

It has rank 1 if $v, w \neq 0$.

3.2.4. Canonical basis. The canonical basis vector $e_i \in \mathbb{R}^n$ has entries $(e_i)_j = \delta_{ij}$, i.e. a 1 in position i and 0 elsewhere. Then $\langle e_i, e_j\rangle = \delta_{ij}$, and every $v \in \mathbb{R}^n$ expands as

$$v = \sum_{i=1}^{n} v_i e_i, \qquad v_i = \langle e_i, v\rangle. \tag{40}$$

3.2.5. Expansion in an orthonormal basis. A set $\{h_1, \ldots, h_n\} \subset \mathbb{R}^n$ is an orthonormal basis (ONB) if $\langle h_i, h_j\rangle = \delta_{ij}$. Orthonormal vectors are linearly independent, so n of them form a basis. Hence $v = \sum_i c_i h_i$ for unique c_i, and $\langle h_j, v\rangle = \sum_i c_i \langle h_j, h_i\rangle = c_j$. Therefore

$$v = \sum_{i=1}^{n} \langle h_i, v\rangle\, h_i, \qquad \langle v, w\rangle = \sum_{i=1}^{n} \langle h_i, v\rangle\langle h_i, w\rangle, \qquad \|v\|^2 = \sum_{i=1}^{n} \langle h_i, v\rangle^2. \tag{41}$$

Example 15. *For $n = 2$, take $h_\pm = (e_1 \pm e_2)/\sqrt{2}$. Then*

$$v = \frac{v_1 + v_2}{\sqrt{2}}\, h_+ + \frac{v_1 - v_2}{\sqrt{2}}\, h_-. \tag{42}$$

3.3. Matrices

3.3.1. Notation. We write

$$A = (a_{ij})_{ij} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{m\times n} \tag{43}$$

for the matrix with entry a_{ij} in row i and column j. The transpose is $A^T = (a_{ji})_{ij}$. It reverses products: $(AB)^T = B^T A^T$.

3.3.2. Canonical basis. The canonical basis of $\mathbb{R}^{m\times n}$ consists of the matrices $E_{ij} = e_i e_j^T$, which have a 1 in position (i, j) and 0 elsewhere. Every $A \in \mathbb{R}^{m\times n}$ expands as

$$A = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} E_{ij}. \tag{44}$$

3.3.3. Matrix–vector multiplication. For $A \in \mathbb{R}^{m\times n}$ and $v \in \mathbb{R}^n$, the product $Av \in \mathbb{R}^m$ has entries

$$(Av)_i = \sum_{j=1}^{n} a_{ij} v_j. \tag{45}$$

Equivalently, $Av = \sum_j v_j\, Ae_j$, a linear combination of the columns Ae_j of A. In particular, $a_{ij} = e_i^T Ae_j$. The rank of A is the dimension of its range $\{Av : v \in \mathbb{R}^n\}$. For an outer product,

$$(vw^T)u = \langle w, u\rangle\, v. \tag{46}$$

3.3.4. Matrix–matrix multiplication. For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, the product $AB \in \mathbb{R}^{m\times p}$ has entries

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}. \tag{47}$$

This is composition: $(AB)v = A(Bv)$. Multiplication is associative and in general not commutative.

3.3.5. Trace. For $A \in \mathbb{R}^{n\times n}$,

$$\operatorname{tr}(A) = \sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} e_i^T A e_i. \tag{48}$$

The trace is linear, and $\operatorname{tr}(A^T) = \operatorname{tr}(A)$.

3.3.6. Cyclicity of the trace. For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$,

$$\operatorname{tr}(AB) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} b_{ji} = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji} a_{ij} = \operatorname{tr}(BA). \tag{49}$$

Consequently, $\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB)$ whenever the products are defined.

Note: In general, $\operatorname{tr}(ABC) \neq \operatorname{tr}(ACB)$. For example, let $A = e_1 e_2^T$, $B = e_2 e_1^T$, $C = e_1 e_1^T$ in $\mathbb{R}^{2\times 2}$. Then

$$\operatorname{tr}(ABC) = \operatorname{tr}(e_1 e_2^T e_2 e_1^T e_1 e_1^T) = \langle e_2, e_2\rangle\langle e_1, e_1\rangle\langle e_1, e_1\rangle = 1, \tag{50}$$

but

$$\operatorname{tr}(ACB) = \operatorname{tr}(e_1 e_2^T e_1 e_1^T e_2 e_1^T) = \langle e_2, e_1\rangle\langle e_1, e_2\rangle\langle e_1, e_1\rangle = 0. \tag{51}$$

3.3.7. Trace inner product. The trace (Hilbert–Schmidt) inner product on $\mathbb{R}^{m\times n}$ is

$$\langle A, B\rangle = \operatorname{tr}(A^T B) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} b_{ij}. \tag{52}$$

It has the same properties as the standard inner product on vectors:

$$\langle A, B\rangle = \langle B, A\rangle \qquad \text{(symmetry)}, \tag{53}$$

$$\langle C, \alpha A + \beta B\rangle = \alpha\langle C, A\rangle + \beta\langle C, B\rangle \qquad \text{(linearity)}, \tag{54}$$

$$\langle A, A\rangle \geq 0, \quad \langle A, A\rangle = 0 \iff A = 0 \qquad \text{(positive definiteness)}. \tag{55}$$

The induced norm is the Frobenius norm $\|A\|_F = \sqrt{\langle A, A\rangle} = \left(\sum_{i,j} a_{ij}^2\right)^{1/2}$.

Remark 16. *Let* $\operatorname{vec}(A) \in \mathbb{R}^{mn}$ *be the vector obtained by stacking the columns of* A*. The trace inner product is the standard inner product of these vectors:*

$$\langle A, B\rangle = \operatorname{tr}(A^T B) = \langle \operatorname{vec}(A), \operatorname{vec}(B)\rangle. \tag{56}$$

3.3.8. Bilinear forms as traces. For $A \in \mathbb{R}^{m\times n}$, $v \in \mathbb{R}^n$ and $w \in \mathbb{R}^m$,

$$w^T A v = \operatorname{tr}(A v w^T). \tag{57}$$

Proof. Using $(vw^T)_{ji} = v_j w_i$,

$$w^T A v = \sum_{i,j} w_i a_{ij} v_j = \sum_{i,j} a_{ij} (vw^T)_{ji} = \sum_{i} (Avw^T)_{ii} = \operatorname{tr}(Avw^T). \quad \square \tag{58}$$

In terms of the two inner products,

$$\langle w, Av\rangle = \langle wv^T, A\rangle. \tag{59}$$

3.3.9. Projectors: trace equals rank. A matrix $P \in \mathbb{R}^{n\times n}$ is an orthogonal projector if $P^2 = P = P^T$. Let U be its range, let $r = \operatorname{rank}(P)$, and let $u_1, \dots, u_r$ be an ONB of U. Then

$$P = \sum_{i=1}^{r} u_i u_i^T, \qquad \operatorname{tr}(P) = \sum_{i=1}^{r} \|u_i\|^2 = r = \operatorname{rank}(P). \tag{60}$$

Proof. Let $Q = \sum_i u_i u_i^T$. If $u = Pw \in U$ for some w, then $Pu = P^2 w = u$, and $Qu = u$ by the ONB expansion. If $x \perp U$, then $\langle Px, y\rangle = \langle x, Py\rangle = 0$ for all y, so $Px = 0$, and also $Qx = 0$. Since $\mathbb{R}^n = U \oplus U^\perp$, we get $P = Q$. The trace formula follows from $\operatorname{tr}(u_i u_i^T) = \langle u_i, u_i\rangle$. □

So on projectors, the rank is the linear function tr. The extension to positive semidefinite matrices is in Section 3.5.

Example 17. *The projector* $h_+ h_+^T = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ *has trace and rank* 1.

3.3.10. Expansion of a matrix in an ONB. Let $\{B_k\}_{k=1}^{mn}$ be an ONB of $\mathbb{R}^{m\times n}$ with respect to the trace inner product, i.e. $\operatorname{tr}(B_k^T B_l) = \delta_{kl}$. Write $A = \sum_k c_k B_k$. As for vectors, taking the inner product with B_l gives $\langle B_l, A\rangle = \sum_k c_k \langle B_l, B_k\rangle = c_l$. Therefore

$$A = \sum_{k=1}^{mn} \langle B_k, A\rangle\, B_k = \sum_{k=1}^{mn} \operatorname{tr}(B_k^T A)\, B_k. \tag{61}$$

Moreover,

$$\|A\|_F^2 = \sum_{k=1}^{mn} \operatorname{tr}(B_k^T A)^2. \tag{62}$$

Canonical basis. The matrices $e_i e_j^T$ form an ONB, and

$$\langle e_i e_j^T, A\rangle = \operatorname{tr}(e_j e_i^T A) = e_i^T A e_j = a_{ij}. \tag{63}$$

Bases induced by an ONB of $\mathbb{R}^n$. If $\{h_i\}$ is an ONB of $\mathbb{R}^n$, then $\{h_i h_j^T\}$ is an ONB of $\mathbb{R}^{n\times n}$, since

$$\langle h_i h_j^T, h_k h_l^T\rangle = \operatorname{tr}(h_j h_i^T h_k h_l^T) = \langle h_i, h_k\rangle\langle h_l, h_j\rangle = \delta_{ik}\delta_{jl}. \tag{64}$$

With $\langle h_i h_j^T, A\rangle = h_i^T A h_j$, this gives

$$A = \sum_{i,j} (h_i^T A h_j)\, h_i h_j^T, \qquad I = \sum_i h_i h_i^T. \tag{65}$$

Example 18 (Pauli matrices and qubits)**.** *In this example only, we work over* $\mathbb{C}$. *The adjoint is* $A^\dagger = \overline{A}^T$, *and the trace inner product on* $\mathbb{C}^{m\times n}$ *is*

$$\langle A, B\rangle = \operatorname{tr}(A^\dagger B) = \sum_{i=1}^{m}\sum_{j=1}^{n} \overline{a_{ij}}\, b_{ij}. \tag{66}$$

It is conjugate symmetric, $\langle A, B\rangle = \overline{\langle B, A\rangle}$, *linear in the second argument, antilinear in the first, and positive definite. The Pauli matrices are*

$$\begin{aligned} \sigma_0 = I &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \sigma_1 = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \\ \sigma_2 = Y &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned} \tag{67}$$

Each σ_a *is Hermitian, i.e.* $\sigma_a^\dagger = \sigma_a$, *and satisfies* $\sigma_a^2 = I$. *Moreover,* $XY = iZ$, $YZ = iX$, $ZX = iY$, *and distinct* X, Y, Z *anticommute. Thus* $\sigma_a\sigma_b$ *is a phase times a traceless Pauli matrix for* $a \neq b$, *so*

$$\langle \sigma_a, \sigma_b\rangle = \operatorname{tr}(\sigma_a\sigma_b) = 2\delta_{ab}. \tag{68}$$

Hence $\{\sigma_a/\sqrt{2}\}_{a=0}^{3}$ *is an ONB of* $\mathbb{C}^{2\times 2}$*, and every* $A \in \mathbb{C}^{2\times 2}$ *expands as*

$$A = \frac{1}{2}\sum_{a=0}^{3} \operatorname{tr}(\sigma_a A)\,\sigma_a. \tag{69}$$

For a general matrix,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{a+d}{2}\,I + \frac{b+c}{2}\,X + \frac{i(b-c)}{2}\,Y + \frac{a-d}{2}\,Z. \tag{70}$$

For example,

$$e_1 e_1^T = \frac{I+Z}{2}, \qquad e_1 e_2^T = \frac{X+iY}{2}, \qquad \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{X+Z}{\sqrt{2}}. \tag{71}$$

If $A = A^\dagger$*, all coefficients* $\operatorname{tr}(\sigma_a A)$ *are real. In particular, a qubit state is a density matrix* $\rho \in \mathbb{C}^{2\times 2}$*, i.e.* $\rho = \rho^\dagger$ *is positive semidefinite (Section 3.5) with* $\operatorname{tr}(\rho) = 1$*. The expansion gives*

$$\rho = \frac{1}{2}\left(I + r_1 X + r_2 Y + r_3 Z\right), \qquad r_a = \operatorname{tr}(\sigma_a \rho) \in \mathbb{R}, \tag{72}$$

where the coefficient of I *is fixed by* $\operatorname{tr}(\rho) = 1$*.*

The matrix $J = iY = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ *is real, with* $J^T = -J$ *and* $J^2 = -I$*. Thus* $\{I, X, J, Z\}/\sqrt{2}$ *is an ONB of the real space* $\mathbb{R}^{2\times 2}$*.*

3.4. Eigenvalues and eigenvectors

3.4.1. Definition. Let $A \in \mathbb{R}^{n\times n}$. A vector $v \neq 0$ is an eigenvector of A with eigenvalue λ if

$$Av = \lambda v. \tag{73}$$

So A acts on the line $\mathbb{R}v$ as multiplication by λ. A number λ is an eigenvalue if and only if $A - \lambda I$ is not invertible, i.e. $\det(A - \lambda I) = 0$. A real matrix need not have real eigenvalues: J is a rotation by $90°$, preserves no line, and has eigenvalues $\pm i$.

3.4.2. Geometric picture. Let $D = \operatorname{diag}(2, \frac{1}{2})$. It maps the unit circle $\{x : \|x\| = 1\}$ onto the ellipse

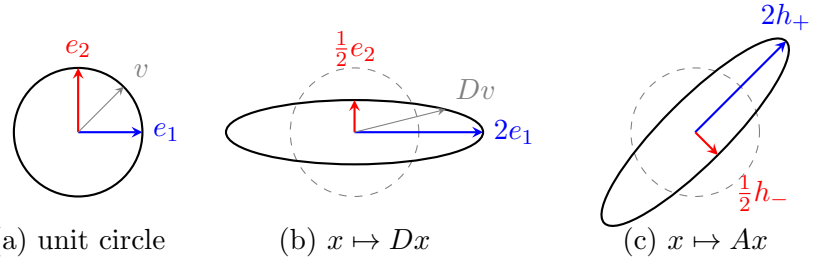
$$\left\{ y \in \mathbb{R}^2 : \frac{y_1^2}{4} + 4y_2^2 = 1 \right\}. \tag{74}$$

The semi-axes of this ellipse lie along e_1, e_2, and their lengths are the eigenvalues $2, \frac{1}{2}$. The eigenvectors are exactly the directions that D only stretches. Any other vector also changes direction, e.g. $v = (1,1)/\sqrt{2}$ maps to $Dv = (2, \frac{1}{2})/\sqrt{2}$.

For a symmetric matrix that is not diagonal, the ellipse is tilted, and its axes are the eigenvectors. For example, let

$$A = \frac{1}{4}\begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}, \qquad Ah_+ = 2h_+, \qquad Ah_- = \frac{1}{2}h_-, \tag{75}$$

with $h_\pm = (e_1 \pm e_2)/\sqrt{2}$. In the ONB $\{h_+, h_-\}$, the matrix A acts as D. Diagonalizing A means choosing the ONB in which the ellipse is axis-aligned. See Figure 1.



(a) unit circle (b) $x \mapsto Dx$ (c) $x \mapsto Ax$

FIGURE 1. Image of the unit circle under a diagonal and a non-diagonal symmetric matrix. The axes of the ellipse are the eigenvectors scaled by the eigenvalues.

3.4.3. Spectral theorem. A matrix $U \in \mathbb{R}^{n\times n}$ is orthogonal if $U^TU = I$, i.e. if its columns form an ONB. Every symmetric matrix $A \in \mathbb{R}^{n\times n}$ has real eigenvalues $\lambda_1, \dots, \lambda_n$ and an ONB of eigenvectors $u_1, \dots, u_n$, so

$$A = U\Lambda U^T = \sum_{i=1}^{n} \lambda_i u_i u_i^T, \qquad U = \begin{pmatrix} | & & | \\ u_1 & \cdots & u_n \\ | & & | \end{pmatrix}, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n). \tag{76}$$

Eigenvectors with different eigenvalues are automatically orthogonal: if $Au = \lambda u$ and $Av = \mu v$, then $\lambda\langle u, v\rangle = \langle Au, v\rangle = \langle u, Av\rangle = \mu\langle u, v\rangle$. The image of the unit sphere is an ellipsoid with semi-axes $\lambda_i u_i$, which is flat if some $\lambda_i = 0$.

3.4.4. Consequences. For symmetric A, cyclicity gives $\operatorname{tr}(A) = \operatorname{tr}(U^TU\Lambda) = \sum_i \lambda_i$, and $\operatorname{rank}(A)$ is the number of nonzero λ_i. Moreover, $x^TAx = \sum_i \lambda_i \langle u_i, x\rangle^2$, so

$$\lambda_{\max}(A) = \max_{\|x\|=1} x^TAx, \qquad \lambda_{\min}(A) = \min_{\|x\|=1} x^TAx. \tag{77}$$

A symmetric matrix is a projector if and only if all its eigenvalues lie in $\{0, 1\}$, since $P^2 = P$ means $\lambda^2 = \lambda$.

3.5. Positive semidefinite matrices

Let $\mathcal{S}^n = \{A \in \mathbb{R}^{n\times n} : A = A^T\}$, a vector space of dimension $n(n+1)/2$ with inner product $\langle A, B\rangle = \operatorname{tr}(AB)$.

3.5.1. Definition. A matrix $A \in \mathcal{S}^n$ is positive semidefinite (PSD), written $A \succeq 0$, if $x^TAx \geq 0$ for all $x \in \mathbb{R}^n$. We write $\mathcal{S}^n_+$ for the set of PSD matrices. Symmetry is part of the definition: $\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ satisfies $x^TAx = \|x\|^2$ but is not PSD.

3.5.2. Theorem (equivalent definitions). For $A \in \mathcal{S}^n$, the following are equivalent:

(a) Quadratic form: $x^TAx \geq 0$ for all $x \in \mathbb{R}^n$.
(b) Eigenvalues: all eigenvalues of A are non-negative.
(c) Sum of rank-one terms: $A = \sum_{l=1}^{m} x_l x_l^T$ for some $x_l \in \mathbb{R}^n$.
(d) Hermitian square: $A = B^TB$ for some $B \in \mathbb{R}^{k\times n}$.
(e) Gram matrix: $a_{ij} = \langle v_i, v_j\rangle$ for some vectors $v_1, \dots, v_n \in \mathbb{R}^k$.
(f) Cholesky factorization: $A = LL^T$ for some lower triangular $L \in \mathbb{R}^{n\times n}$.
(g) Self-duality: $\operatorname{tr}(AB) \geq 0$ for all $B \in \mathcal{S}^n_+$.

Proof. (a) $\Rightarrow$ (b): If $Au = \lambda u$ with $\|u\| = 1$, then $\lambda = u^TAu \geq 0$.
(b) $\Rightarrow$ (c): Take $x_i = \sqrt{\lambda_i}\, u_i$ in the spectral decomposition.
(c) $\Rightarrow$ (d): Let B have rows x_l^T. Then $B^TB = \sum_l x_l x_l^T$.
(d) $\Rightarrow$ (e): Let v_j be the columns of B. Then $(B^TB)_{ij} = \langle v_i, v_j\rangle$.
(e) $\Rightarrow$ (a): $x^TAx = \sum_{i,j} x_i x_j \langle v_i, v_j\rangle = \|\sum_i x_i v_i\|^2 \geq 0$.

(b) ⇒ (f): The matrix $B = \Lambda^{1/2}U^T$ is square with $B^TB = A$. Take a QR decomposition $B = QR$, with Q orthogonal and R upper triangular. Then $A = R^TQ^TQR = R^TR$, so $L = R^T$ works.

(f) ⇒ (d): Take $B = L^T$.

(a) ⇒ (g): By (c), any $B \in \mathcal{S}^n_+$ can be written as $B = \sum_l x_l x_l^T$. Then $\operatorname{tr}(AB) = \sum_l x_l^T A x_l \geq 0$.

(g) ⇒ (a): The matrix xx^T is PSD by (c), and $\operatorname{tr}(Axx^T) = x^TAx$. □

3.5.3. Remarks. The matrix A is positive definite, written $A \succ 0$, if $x^TAx > 0$ for all $x \neq 0$. This is equivalent to all eigenvalues being positive, and to B in (d) having full column rank.

Example 19. *The matrix* $\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$ *is PSD, with eigenvalues* $2, 0$. *The matrix* $\left(\begin{smallmatrix}1&2\\2&1\end{smallmatrix}\right)$ *is not, since* $x = (1,-1)$ *gives* $x^TAx = -2$.

Trace as a proxy for rank. Let $A \succeq 0$ with $A \neq 0$. Since $0 \leq \lambda_i \leq \lambda_{\max}(A)$,

$$\frac{\operatorname{tr}(A)}{\lambda_{\max}(A)} \leq \operatorname{rank}(A), \tag{78}$$

with equality if and only if A is a multiple of a projector. If all eigenvalues of A are at most 1, this gives $\operatorname{tr}(A) \leq \operatorname{rank}(A)$. This is why the rank is often replaced by the trace in semidefinite relaxations.

Example 20 (Qubits)**.** *Over* $\mathbb{C}$, *a matrix is PSD if* $A = A^\dagger$ *and* $x^\dagger Ax \geq 0$ *for all* $x \in \mathbb{C}^n$. *The matrix* $r_1X + r_2Y + r_3Z$ *is traceless and squares to* $\|r\|^2 I$, *so it has eigenvalues* $\pm\|r\|$. *Thus* ρ *has eigenvalues* $(1 \pm \|r\|)/2$, *and* $\rho \succeq 0$ *if and only if* $\|r\| \leq 1$. *Qubit states form the Bloch ball.*

3.6. Convex sets and cones

3.6.1. Löwner order. For $A, B \in \mathcal{S}^n$, we write $A \succeq B$ if $A - B \succeq 0$. This is a partial order: it is reflexive, transitive, and antisymmetric, since $A \succeq B \succeq A$ gives $\lambda_i(A-B) = 0$ for all i, hence $A = B$. It is not a total order: there are matrices A, B that are incomparable, meaning that neither $A \succeq B$ nor $B \succeq A$ holds. Can you find two incomparable 2×2 matrices?

Cones, convex sets, and convex hulls are defined in Section 1.2.

3.6.2. The PSD cone. The set $\mathcal{S}^n_+$ is a convex cone: if $A, B \succeq 0$ and $\alpha, \beta \geq 0$, then

$$x^T(\alpha A + \beta B)x = \alpha\, x^TAx + \beta\, x^TBx \geq 0. \tag{79}$$

It is closed, since it is an intersection of the closed half-spaces $\{A : \langle xx^T, A\rangle \geq 0\}$. By (c), it is the set of finite sums of the rank-one matrices xx^T. The set of PSD matrices of rank at most 1 is a cone, but it is not convex for $n \geq 2$. The set of PSD matrices of trace 1 is the convex hull of the rank-one projectors:

$$\{A \in \mathcal{S}^n_+ : \operatorname{tr}(A) = 1\} = \operatorname{conv}\{xx^T : \|x\| = 1\}. \tag{80}$$

Here "$\supseteq$" holds since the left-hand side is convex and contains each xx^T. For "$\subseteq$", write $A = \sum_i \lambda_i u_i u_i^T$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i = \operatorname{tr}(A) = 1$. This set is the real analogue of the set of density matrices.

For $n = 2$, the matrix $A = \left(\begin{smallmatrix}a&b\\b&c\end{smallmatrix}\right)$ has eigenvalues $\frac{1}{2}\big(a + c \pm \sqrt{(a-c)^2 + 4b^2}\big)$. Thus $A \succeq 0$ if and only if $a + c \geq \sqrt{(a-c)^2 + 4b^2}$. In the orthonormal coordinates $(a+c, a-c, 2b)/\sqrt{2}$, this is a circular cone in $\mathbb{R}^3$. Its boundary consists of the matrices of rank at most 1. See Figure 2(b).

3.6.3. Dual cone. Let V be a vector space with an inner product, and let $K \subseteq V$ be a cone. The dual cone is

$$K^* = \{y \in V : \langle y, x\rangle \geq 0 \text{ for all } x \in K\}. \tag{81}$$

It is always a closed convex cone, as an intersection of closed half-spaces. If K is a closed convex cone, then $K^{**} = K$. A cone is self-dual if $K^* = K$.

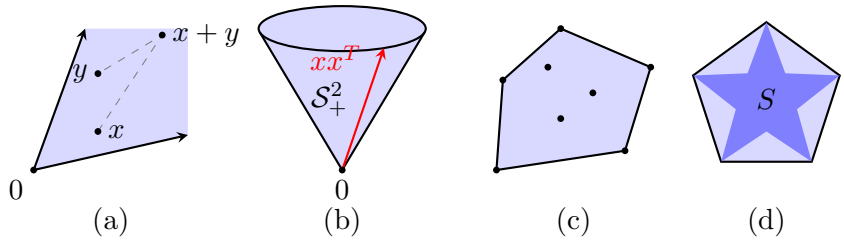


FIGURE 2. (a) A convex cone, which contains $x + y$ for all its points x, y. (b) The cone $\mathcal{S}^2_+$, with a boundary ray generated by a rank-one matrix xx^T. (c) The convex hull of a finite set of points. (d) The convex hull (light) of a non-convex set S (dark).

Example 21. *The orthant $\mathbb{R}^n_+$ is self-dual: if $\langle y, x\rangle \geq 0$ for all $x \geq 0$, then $y_i = \langle y, e_i\rangle \geq 0$, and the converse is clear. The ray $K = \{\alpha e_1 : \alpha \geq 0\} \subset \mathbb{R}^2$ is not self-dual: K^* is the half-plane $\{y : y_1 \geq 0\}$.*

3.6.4. Self-duality of the PSD cone. In $\mathcal{S}^n$ with the trace inner product,

$$(\mathcal{S}^n_+)^* = \mathcal{S}^n_+. \tag{82}$$

This is the equivalence (a) $\Leftrightarrow$ (g). The dual cone depends on the ambient space: in $\mathbb{R}^{n\times n}$, the dual of $\mathcal{S}^n_+$ is $\{A : A + A^T \succeq 0\}$, which contains all antisymmetric matrices.

As a consequence, $A \succeq B$ if and only if $\operatorname{tr}(AC) \geq \operatorname{tr}(BC)$ for all $C \succeq 0$.

CHAPTER 4

Reading list

i) L. Lovász, *Semidefinite programs and combinatorial optimization*, lecture notes.
https://sites.math.washington.edu/~thomas/teaching/m514_web/Lovasz_semidef.pdf

ii) A. Gupta, R. O'Donnell, *Linear and semidefinite programming*, lecture notes, CMU course 15-859, Fall 2011. Uses the same convention for primal and dual as these notes.
https://www.cs.cmu.edu/afs/cs.cmu.edu/academic/class/15859-f11/www/notes/lpsdp.pdf

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https://link.springer.com/content/pdf/10.1007/978-3-540-30717-4.pdf

iv) H. Yang, *Semidefinite Optimization and Relaxation*, lecture notes. Uses a different convention for primal and dual than these notes.
https://hankyang.seas.harvard.edu/Semidefinite/

v) G. Blekherman, P. A. Parrilo, R. R. Thomas (eds.), *Semidefinite Optimization and Convex Algebraic Geometry*, SIAM, 2013.
https://www.mit.edu/~parrilo/sdocag/

vi) *JuMP*, a modeling language for mathematical optimization in Julia.
https://jump.dev/

APPENDIX A

Proof of Farkas Lemma

The aim of this appendix is to prove the direction of Theorem 3 that is not proven in Section 1.3: if $Ax \le b$ has no solution, then there is $\lambda \ge 0$ with $A^T\lambda = \mathbf{0}$ and $b^T\lambda = -1$. We use Fourier–Motzkin elimination, following [11, Ch. 1]; see also [9]. The idea is to remove the variables $x_1, \dots, x_n$ one after another by taking non-negative combinations of the inequalities.

Lemma 22 (Fourier–Motzkin elimination step)**.** *Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and $k \in \{1, \dots, n\}$. Suppose that the columns $1, \dots, k-1$ of A are zero. Then there exist $m' \ge 0$ and a matrix $M \in \mathbb{R}^{m'\times m}$ with $M \ge 0$ such that*

(a) the columns $1, \dots, k$ of $A' := MA$ are zero, and
(b) with $b' := Mb$, the system $Ax \le b$ has a solution if and only if the system $A'x \le b'$ has a solution.

(If $m' = 0$, the system $A'x \le b'$ is empty, and every $x \in \mathbb{R}^n$ is a solution.)

Proof. Denote the entries of A by a_{ij}. Split the row indices into

$$P = \{i : a_{ik} > 0\}\,, \quad N = \{i : a_{ik} < 0\}\,, \quad Z = \{i : a_{ik} = 0\}\,. \tag{83}$$

The matrix M has one row for each $i \in Z$ and one row for each pair $(i,j) \in P \times N$:

$$\text{row } i \in Z\text{: } e_i^T\,, \qquad \text{row } (i,j) \in P\times N\text{: } \frac{1}{a_{ik}}e_i^T + \frac{1}{|a_{jk}|}e_j^T\,. \tag{84}$$

All entries of M are non-negative.

(a) Row $i \in Z$ of MA is a_i^T, whose k-th entry is $a_{ik} = 0$. Row (i,j) of MA is $a_i^T/a_{ik} + a_j^T/|a_{jk}|$, whose k-th entry is $1 - 1 = 0$. The entries $1, \dots, k-1$ of these rows are zero, because those of a_i and a_j are.

(b) If $Ax \le b$, then $MAx \le Mb$ by (2), applied to each row of M. So every solution of $Ax \le b$ solves $A'x \le b'$.

Conversely, let x solve $A'x \le b'$. We keep all entries of x except x_k, and choose a new value t for x_k. Let $x(t)$ be the vector x with its k-th entry replaced by t. Since the k-th column of A' is zero, $x(t)$ solves $A'x(t) \le b'$ for every $t \in \mathbb{R}$. Write $a_i^T x(t) = a_{ik}t + r_i$, where $r_i = a_i^T x - a_{ik}x_k$ does not depend on t. The i-th inequality of $Ax(t) \le b$ reads

$$\begin{aligned} i \in P\text{:}\quad & t \le u_i := \frac{b_i - r_i}{a_{ik}}\,, \\ i \in N\text{:}\quad & t \ge l_i := \frac{b_i - r_i}{a_{ik}}\,, \\ i \in Z\text{:}\quad & r_i \le b_i\,. \end{aligned} \tag{85}$$

For $i \in Z$, the inequality $r_i \le b_i$ is row i of $A'x \le b'$, so it holds. For $(i,j) \in P \times N$, row (i,j) of $A'x(t) \le b'$ reads

$$\frac{a_{ik}t + r_i}{a_{ik}} + \frac{a_{jk}t + r_j}{|a_{jk}|} \le \frac{b_i}{a_{ik}} + \frac{b_j}{|a_{jk}|}\,. \tag{86}$$

The terms with t cancel, and using $|a_{jk}| = -a_{jk}$ this is equivalent to $l_j \le u_i$. Thus $l_j \le u_i$ for all $i \in P$ and $j \in N$, and therefore

$$\max_{j\in N} l_j \le \min_{i\in P} u_i\,, \tag{87}$$

where the maximum over the empty set is $-\infty$ and the minimum over the empty set is $+\infty$. Choose t between these two values. Then $x(t)$ satisfies all inequalities of $Ax \le b$. □

PROOF OF THEOREM 3: IF I) FAILS, THEN II) HOLDS. Suppose that $Ax \le b$ has no solution. Apply Lemma 22 with $k = 1$ to (A, b), then with $k = 2$ to the resulting system, and so on up to $k = n$. This gives matrices $M_1, \dots, M_n$ with non-negative entries. Their product $M = M_n \cdots M_1$ also has non-negative entries. After the last step all columns are zero, so $MA = 0$. Let $b^{(n)} = Mb$. By part (b) of Lemma 22, applied in each step, the system $0 \cdot x \le b^{(n)}$ has no solution. This system consists of the inequalities $0 \le b^{(n)}_\ell$. It has no solution, so it is not empty and $b^{(n)}_\ell < 0$ for some row ℓ. Let μ^T be row ℓ of M. Then $\mu \ge 0$, $A^T\mu = (\mu^T A)^T = \mathbf{0}$ (row ℓ of $MA = 0$), and $b^T\mu = b^{(n)}_\ell < 0$. The vector $\lambda = \mu/|b^T\mu|$ satisfies (3). □

Example 23. *Consider the system*

(i) $$x_1 + x_2 \le 1$$

(ii) $$-x_1 \le 0$$

(iii) $$-x_2 \le -2\,.$$

It has no solution: (ii) and (iii) say $x_1 \ge 0$ and $x_2 \ge 2$, so $x_1 + x_2 \ge 2$. Eliminate x_1: here $P = \{i\}$, $N = \{ii\}$, $Z = \{iii\}$. The new system is

$$\begin{aligned} x_2 &\le 1 \qquad \textit{(i)+(ii)} \\ -x_2 &\le -2 \qquad \textit{(iii)}\,. \end{aligned} \tag{88}$$

Eliminate x_2: the sum of the two inequalities is $0 \le -1$. It is the sum of (i), (ii), (iii). So $\lambda = (1, 1, 1)^T$, and indeed $A^T\lambda = (1 - 1 + 0, 1 + 0 - 1)^T = \mathbf{0}$ and $b^T\lambda = 1 + 0 - 2 = -1$.

Remark 24. *Fourier–Motzkin elimination is also an algorithm to decide whether $Ax \le b$ has a solution. It is not efficient: one step can replace m inequalities by up to $|P| \cdot |N| \le m^2/4$ inequalities.*

To summarize this appendix: eliminating all variables by non-negative combinations of the inequalities turns an infeasible system $Ax \le b$ into an inequality $0 \le b^T\mu$ with $b^T\mu < 0$, and the weights μ, rescaled, are the vector λ of Farkas Lemma.

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